

Git Hands-on Lab 1 Report – Git Installation, Configuration and First Commit

To install Git, configure the Git environment, create a local Git repository, add a file to the repository, commit the changes, and verify the repository status.

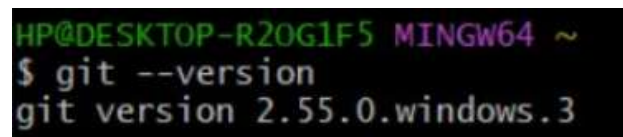
Prerequisites

- Windows Operating System
- Git for Windows installed
- Notepad++ installed
- Git Bash

Step 1: Verify Git Installation

Open Git Bash and verify that Git is installed successfully.

```
git --version
```



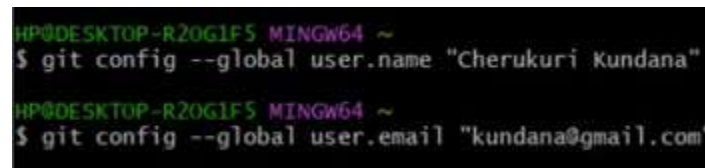
```
HP@DESKTOP-R2OG1F5 MINGW64 ~  
$ git --version  
git version 2.55.0.windows.3
```

Step 2: Configure Git User Information

Configure the Git username and email address.

```
git config --global user.name "Cherukuri Kundana"
```

```
git config --global user.email "kundana@gmail.com"
```



```
HP@DESKTOP-R2OG1F5 MINGW64 ~  
$ git config --global user.name "Cherukuri Kundana"  
HP@DESKTOP-R2OG1F5 MINGW64 ~  
$ git config --global user.email "kundana@gmail.com"
```

Step 3: Configure Notepad++ as the Default Editor

Set Notepad++ as the default editor for Git.

```
git config --global core.editor "C:/Program Files/Notepad++/notepad++.exe"
```



```
HP@DESKTOP-R2OG1F5 MINGW64 ~  
$ git config --global core.editor "C:/Program Files/Notepad++/notepad++.exe"
```

Step 4: Verify Git Configuration

Verify all the configured Git settings.

git config --list

```
HP@DESKTOP-R20G1F5 MINGW64 ~  
$ git config --list  
diff.astextplain.textconv=astextplain  
filter.lfs.clean=git-lfs clean -- %f  
filter.lfs.smudge=git-lfs smudge -- %f  
filter.lfs.process=git-lfs filter-process  
filter.lfs.required=true  
http.sslbackend=schannel  
core.autocrlf=true  
core.fscache=true  
core.symlinks=false  
pull.rebase=false  
credential.helper=manager  
credential.https://dev.azure.com.usehttppath=true  
init.defaultbranch=master  
user.name=Cherukuri Kundana  
user.email=kundana@gmail.com  
core.editor=C:/Program Files/Notepad++/notepad++.exe
```

Step 5: Create a Local Repository

Create a new folder named **GitLab**, navigate to it, and initialize a Git repository.

mkdir GitLab

cd GitLab

git init

```
HP@DESKTOP-R20G1F5 MINGW64 ~  
$ mkdir GitLab  
  
HP@DESKTOP-R20G1F5 MINGW64 ~  
$ cd GitLab  
  
HP@DESKTOP-R20G1F5 MINGW64 ~/GitLab  
$ git init  
Initialized empty Git repository in C:/Users/HP/GitLab/.git/
```

Step 6: Check Repository Status

Verify the status of the newly created repository.

git status

```
HP@DESKTOP-R20G1F5 MINGW64 ~/GitLab (master)  
$ git status  
On branch master  
  
No commits yet  
  
nothing to commit (create/copy files and use "git add" to track)
```

Step 7: Create a New File

Create a text file named **hello.txt**.

echo "This is my first Git file." > hello.txt

```
HP@DESKTOP-R20G1F5 MINGW64 ~/GitLab (master)
$ echo "This is my first Git file." > hello.txt
```

Step 8: Verify Repository Status

Check the repository status after creating the file.

git status

```
HP@DESKTOP-R20G1F5 MINGW64 ~/GitLab (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Step 9: Stage the File

Add **hello.txt** to the staging area.

git add hello.txt

```
HP@DESKTOP-R20G1F5 MINGW64 ~/GitLab (master)
$ git add hello.txt
warning: in the working copy of 'hello.txt', LF will be replaced by CRLF the next time Git touches it
```

Step 10: Verify the Staging Area

Check that the file has been staged successfully.

git status

```
HP@DESKTOP-R20G1F5 MINGW64 ~/GitLab (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   hello.txt
```

Step 11: Commit the Changes

Commit the staged file to the repository.

git commit -m "Added hello.txt"

```
HP@DESKTOP-R2OG1F5 MINGW64 ~/GitLab (master)
$ git commit -m "Added hello.txt"
[master (root-commit) 15f4d2e] Added hello.txt
1 file changed, 1 insertion(+)
create mode 100644 hello.txt
```

Step 12: Final Verification

Verify that the repository is in a clean working state.

git status

```
HP@DESKTOP-R2OG1F5 MINGW64 ~/GitLab (master)
$ git status
On branch master
nothing to commit, working tree clean
```

Result

Git was installed and configured successfully. A local Git repository named **GitLab** was created, and a text file named **hello.txt** was added, staged, and committed successfully. The repository status was verified, and the working tree was clean with the message "**nothing to commit, working tree clean.**"